Homework 3

1. The demand for a type of carpet known as KP-7 has been estimated as:

   P = 40 -0.25Q where P is price and Q is quantity.

   The market supply is expressed as:

   P = 5 +0.05 Q.

   A typical firm in this market has a total cost function given as:

   $TC = 100 - 20q + 2q^2$

   a. Determine the equilibrium market output and price.
   b. Determine the output for a typical firm.
   c. Determine the profit (or loss) earned by the typical firm.

2. The handmade snuffbox industry is composed of 100 identical firms, each having short-run total costs given by $STC = 0.5q^2 + 10q + 5$, where q is the output of snuffboxes per day.
   a. What is the short-run supply curve for each snuffbox maker?
   b. What is the short-run supply curve for the market as a whole?
   c. Suppose the demand for total snuffbox production is given by $Q = 1100 - 50P$. What is the equilibrium in this marketplace? What is each firm's total short-run profit?
   d. Graph the market equilibrium and compute total producer surplus in this case.
   e. Suppose now that the government imposed a $3 tax on snuffboxes. How would this tax change the market equilibrium?
   f. Calculate the total loss of producer surplus as a result of taxation of snuffboxes. Show that this loss equals the change in total short-run profits in the snuffbox industry. Why don't fixed costs enter into this computation of the change in short-run producer surplus?